



Chapter 21

Network Layer: Address Mapping,

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21-1 ADDRESS MAPPING

*The delivery of a packet to a host or a router requires two levels of addressing: **logical** and **physical**. We need to be able to map a logical address to its corresponding physical address and vice versa. This can be done by using either static or dynamic mapping.*

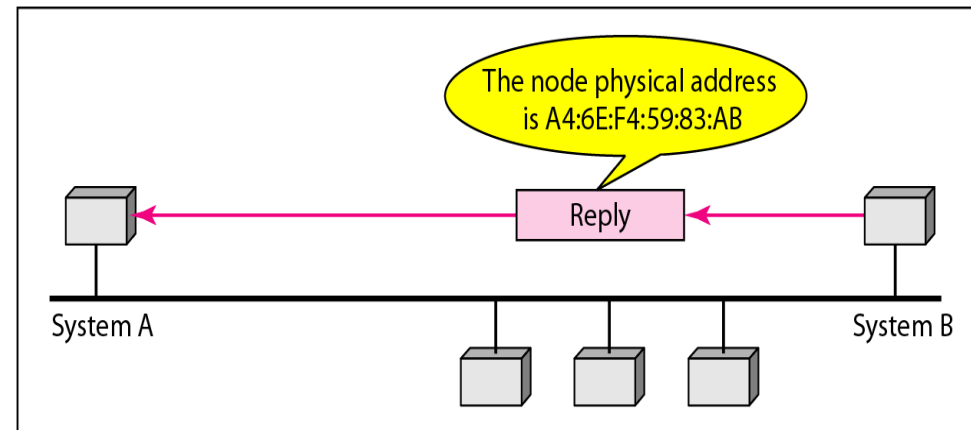
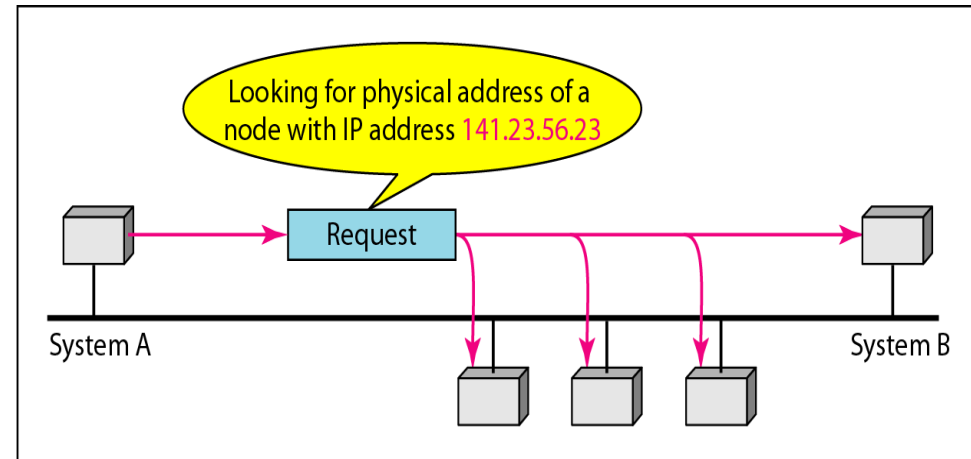
Topics discussed in this section:

Mapping Logical to Physical Address

Mapping Physical to Logical Address

Figure 21.1 ARP operation

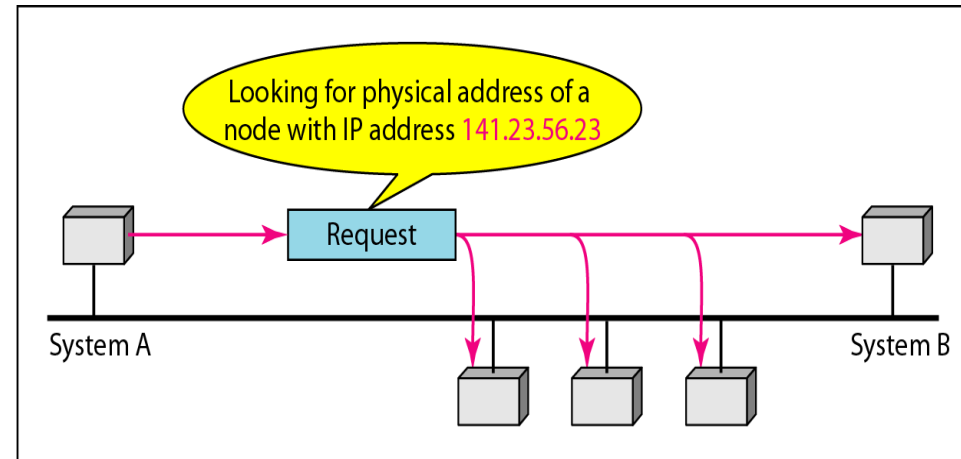
- The host or the router sends an ARP query packet. The packet includes the physical and IP addresses of the sender and the IP address of the receiver.
- The sender does not know the physical address of the receiver, the query is broadcast over the network shown in fig. a.
- Every host or router on the network receives and processes the ARP query packet, but only the intended recipient recognizes its IP address and sends back an ARP response packet.
- The response packet contains the recipient's IP and physical addresses.
- The packet is unicast directly to the inquirer by using the physical address received in the query packet.



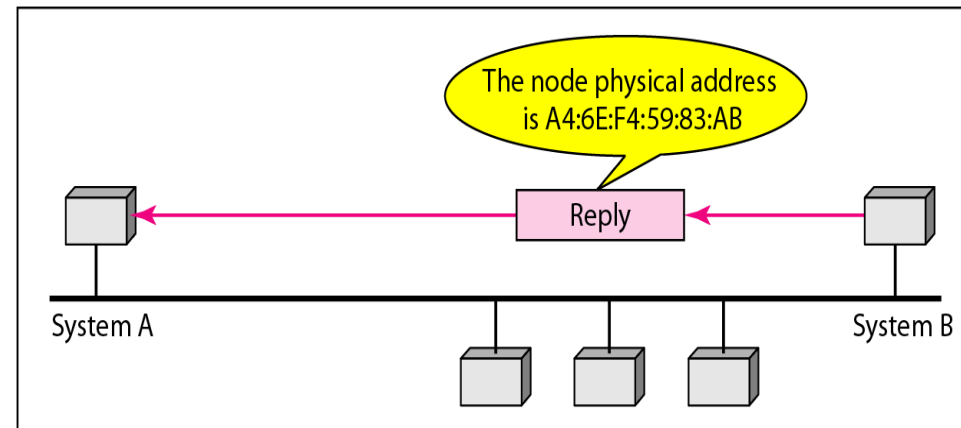
b. ARP reply is unicast

Figure 21.1 ARP operation

- In Figure 21.1a, the system on the left (A) has a packet that needs to be delivered to another system (B) with IP address 141.23.56.23.
- System A needs to pass the packet to its data link layer for the actual delivery, but it does not know the physical address of the recipient.
- It uses the services of ARP by asking the ARP protocol to send a broadcast ARP request packet to ask for the physical address of a system with an IP address of 141.23.56.23.

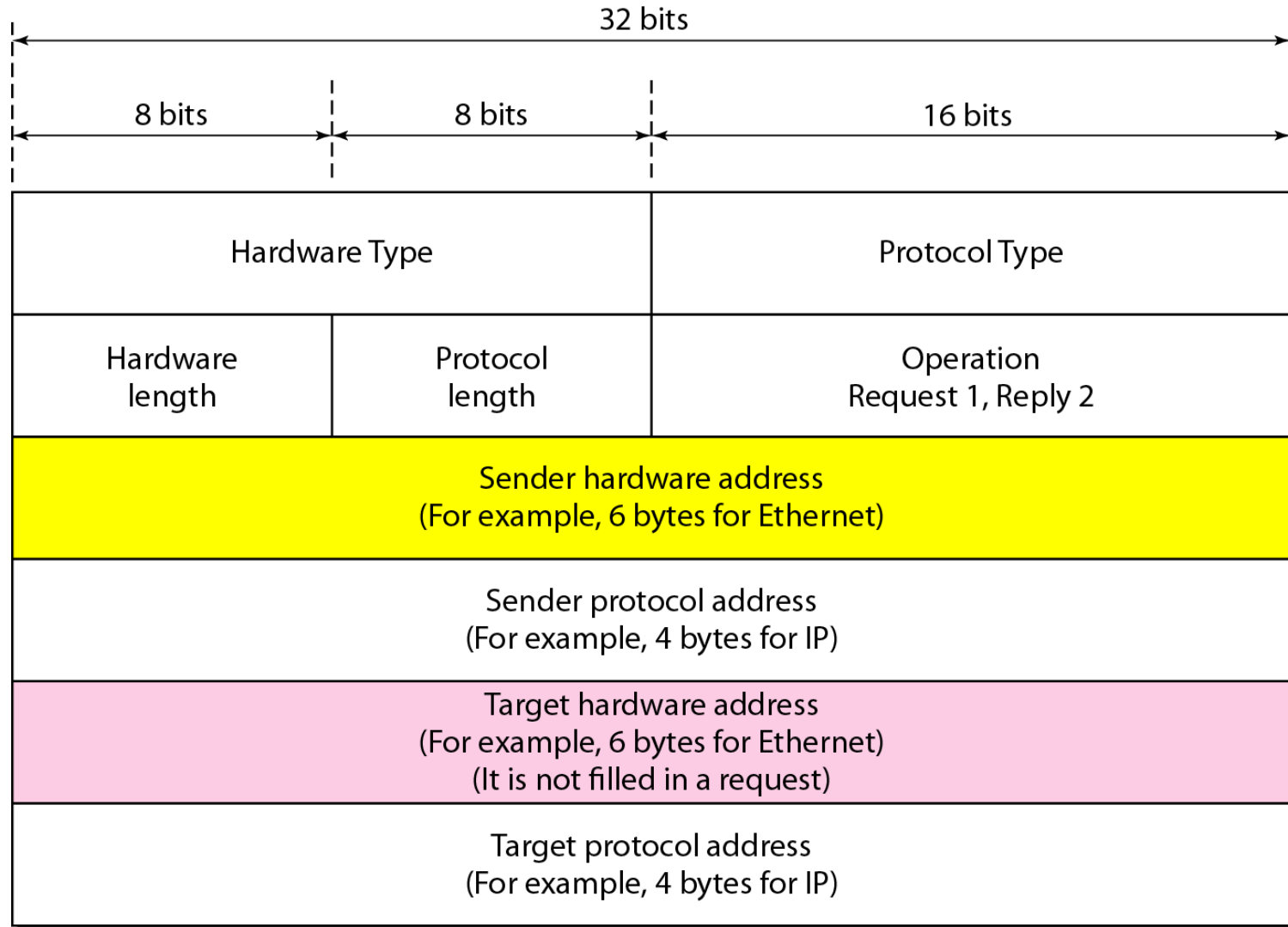


a. ARP request is broadcast



b. ARP reply is unicast

Figure 21.2 *ARP packet*



ARP Packet Fields

- *The fields are as follows:*
- **Hardware type:** This is a 16-bit field defining the type of the network on which ARP is running. For example, Ethernet is given type 1. ARP can be used on any physical network.
- **Protocol type:** This is a 16-bit field defining the protocol. For example, the value of this field for the IPv4 protocol is 0800_{16} .
- **Hardware length:** This is an 8-bit field defining the length of the physical address in bytes. For example, for Ethernet, the value is 6.
- **Protocol length:** This is an 8-bit field defining the length of the logical address in bytes. For example, for the IPv4 protocol, the value is 4.
- **Operation:** This is a 16-bit field defining the type of packet. Two packet types are defined: ARP request (1) and ARP reply (2).
- **Sender hardware address:** This is a variable-length field defining the physical address of the sender. For example, for Ethernet, this field is 6 bytes long.

ARP Packet Fields(Con..

- *The fields are as follows:*
- **Sender protocol address:** *This is a variable-length field defining the logical (for example, IP) address of the sender. For the IP protocol, this field is 4 bytes long.*
- **Target hardware address:** *This is a variable-length field defining the physical address of the target. For example, for Ethernet, this field is 6 bytes long.*
- **Target protocol address:** *This is a variable-length field defining the logical (for example, IP) address of the target. For the IPv4 protocol, this field is 4 bytes long.*



Note

**An ARP request is broadcast;
an ARP reply is unicast.**



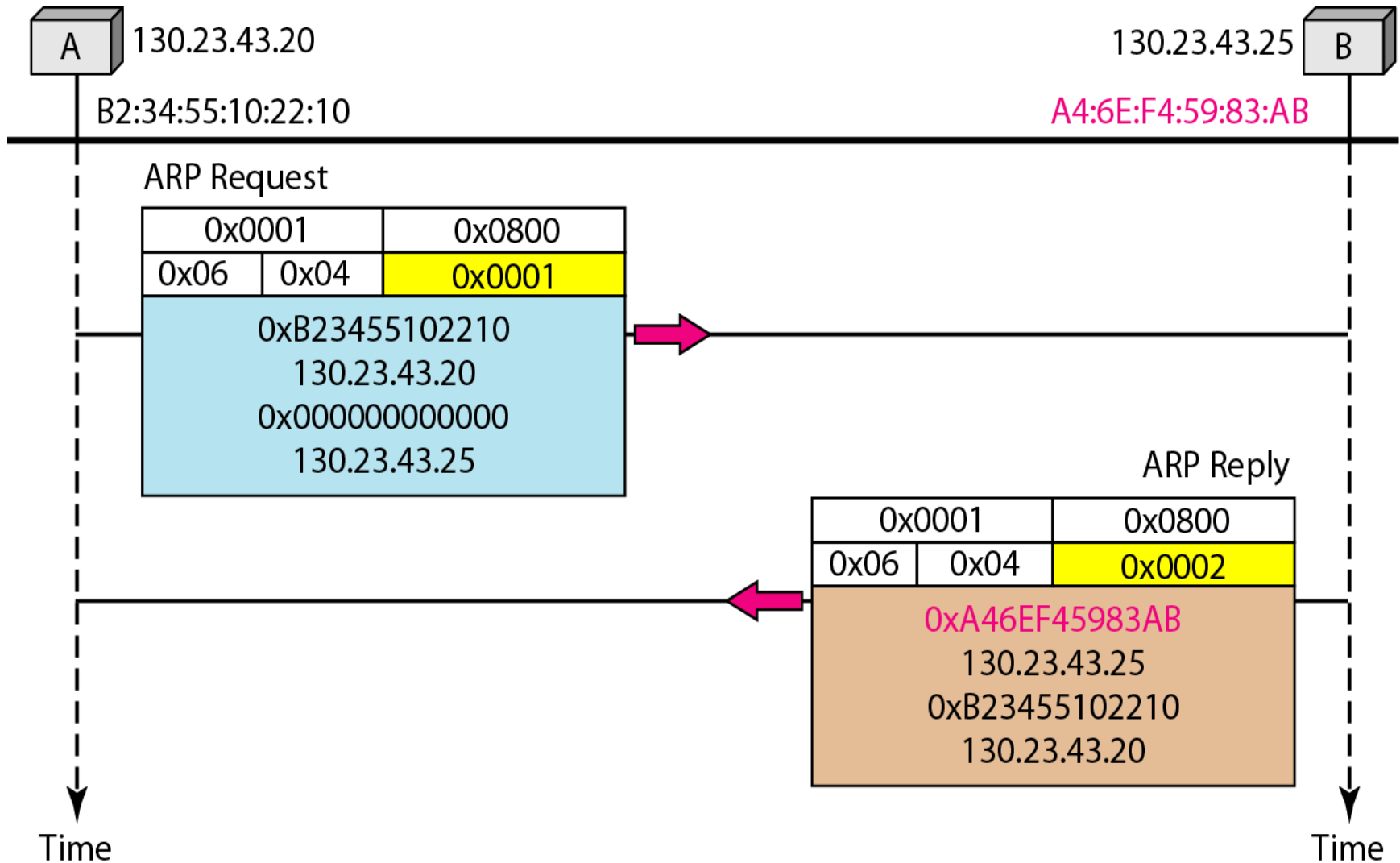
Example 21.1

A host with IP address 130.23.43.20 and physical address B2:34:55:10:22:10 has a packet to send to another host with IP address 130.23.43.25 and physical address A4:6E:F4:59:83:AB. The two hosts are on the same Ethernet network. Show the ARP request and reply packets encapsulated in Ethernet frames.

Solution

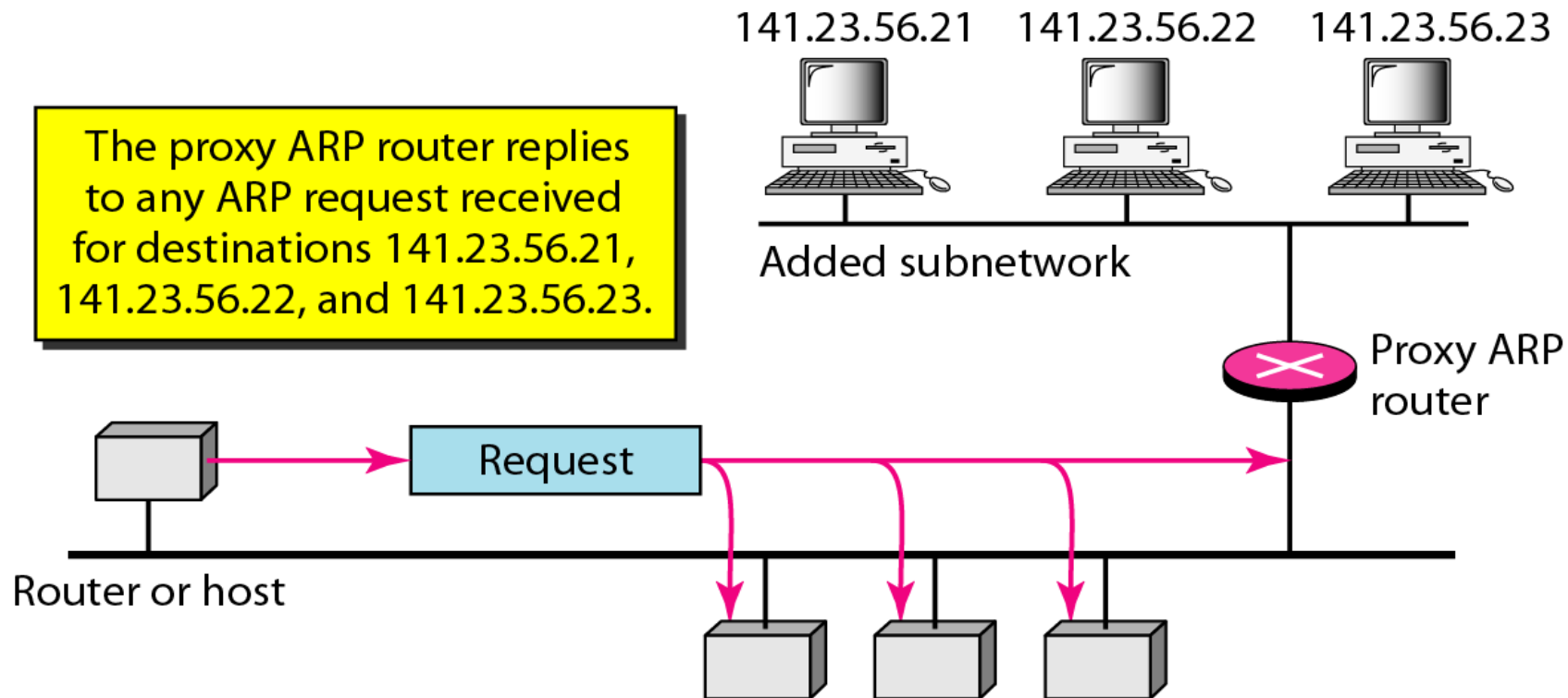
Figure 21.5 shows the ARP request and reply packets. Note that the ARP data field in this case is 28 bytes, and that the individual addresses do not fit in the 4-byte boundary. That is why we do not show the regular 4-byte boundaries for these addresses.

ARP REQUEST AND REPLY



PROXY ARP

- A proxy ARP is an ARP that acts on behalf of a set of hosts.
- When a proxy ARP router receives an ARP request for another host, it replies with its own MAC (hardware) address.
- After the router receives the actual IP packet, it sends the packet to the appropriate host or router.



PROXY ARP(con..)

- In Figure 21.6 the ARP installed on the right-hand host will answer only to an ARP request with a target IP address of 141.23.56.23.
- When it receives an ARP request for one of its protected hosts (141.23.56.21, 141.23.56.22, or 141.23.56.23), it replies with its own MAC (hardware) address.
- When the router receives the IP packet, it sends the packet to the appropriate host.

RARP conversation



The diagram illustrates a Reverse Address Resolution Protocol (RARP) conversation. A horizontal black line represents the network bus. On the left, a green square represents the requester. A green arrow points from this square to a green-bordered box containing the request message. On the right, a blue square represents the responder. A blue arrow points from this square to a blue-bordered box containing the response message. Between the two nodes, four dotted orange arrows point towards the text 'not me', indicating that the other three nodes on the bus did not respond to the request.

HEY - Everyone please listen!
My Ethernet address is
22:BC:66:17:01:75.
Does anyone know my IP address ?

not me

Hi Green! Your IP address is
128.213.1.17.

21-2 ICMP

*The IP protocol has no error-reporting or error-correcting mechanism. The IP protocol also lacks a mechanism for host and management queries. The **Internet Control Message Protocol (ICMP)** has been designed to compensate for the above two deficiencies. It is a companion to the IP protocol.*

Topics discussed in this section:

Types of Messages

Message Format

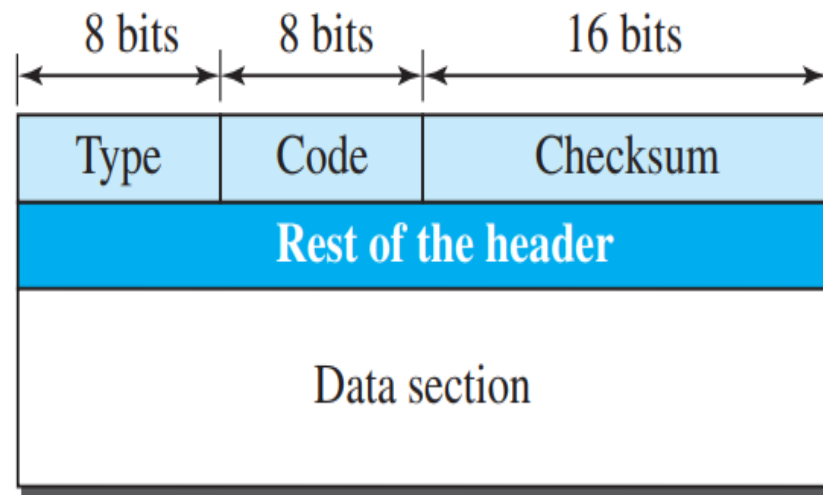
Error Reporting and Query

Debugging Tools

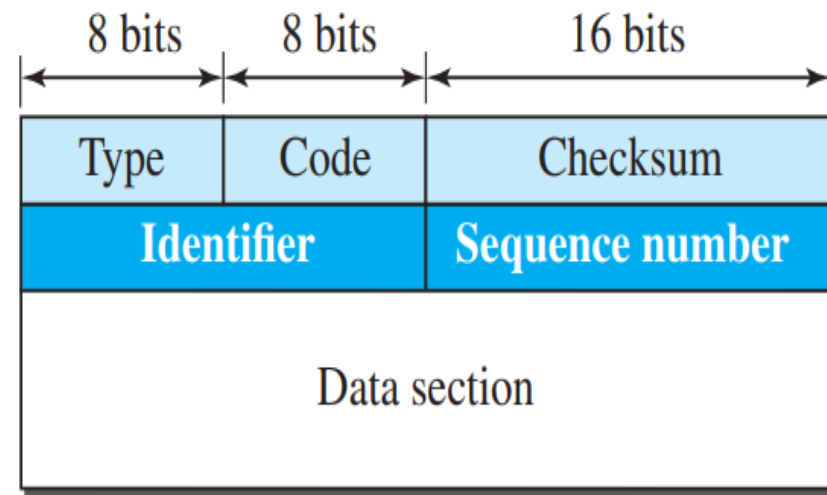
ICMP: Internet Control Message Protocol

- *ICMP is a protocol used for exchanging control messages.*
- *ICMP uses IP to deliver messages.*
- *IPv4 provides no built-in error reporting, error correction, or network management.*
- *These functions are handled by ICMP, which reports errors and supports diagnostic and control messages.*

ICMP messages



Error-reporting messages



Query messages

- *ICMP messages are divided into error-reporting messages and query messages.*
- *The error-reporting messages report problems that a router or a host (destination) may encounter.*
- *The query messages get specific information from a router or another host.*

ICMP messages

<i>Category</i>	<i>Type</i>	<i>Message</i>
Error-reporting messages	3	Destination unreachable
	4	Source quench
	11	Time exceeded
	12	Parameter problem
	5	Redirection
Query messages	8 or 0	Echo request or reply
	13 or 14	Timestamp request or reply
	17 or 18	Address mask request or reply
	10 or 9	Router solicitation or advertisement

ICMP: Error Reporting Messages

- *Destination Unreachable (ICMP Type 3): This ICMP error message is sent when an IP packet cannot reach its destination.*
- *Type 3 uses different codes (0–15) to indicate the specific reason, such as network unreachable or host unreachable.*
[Assignment on full list of codes]
- *Source Quench (ICMP Type 4): This message informs the sender that a packet was dropped due to network congestion and requests the sender to reduce its transmission rate.*
- *Redirect Message (ICMP Type 5): This message is sent when a host uses the wrong router. The router forwards the packet through the correct router and informs the sender to use that router for future packets.*

ICMP: Error Reporting Messages

- *Time Exceeded (ICMP Type 11): This message is sent when a packet is discarded because*
 - *TTL reaches 0 (Code 0) or*
 - *A packet do not arrive within the reassembly time limit (Code 1).*
- *Parameter Problem (ICMP Type 12): This message is sent when an IP packet contains an*
 - *invalid header field (Code 0) or*
 - *has missing or unrecognized options (Code 1).*

ICMP messages

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ICMP: Query Messages

- *ICMP Query Messages used to test network connectivity, measure delay, and check host or router availability through request–reply messages in pairs.*
- *Echo Request (ICMP Type 8) and Echo Reply (Type 0): These messages are used to test whether a host or router is reachable; the destination replies with an Echo Reply if it is active (used by the ping utility).*
- *Router Advertisement (ICMP Type 9) and Router Solicitation (Type 10): These messages enable hosts to discover available routers; a host sends a Router Solicitation (Type 10), and a router responds with a Router Advertisement (Type 9).*

ICMP: Query Messages

- *Timestamp Request (ICMP Type 13) and Timestamp Reply (Type 14): These messages are used to measure the round-trip time (RTT) and check clock synchronization between two devices.*
- *Address Mask Request (ICMP Type 17) and Address Mask Reply (Type 18): These messages allow a host to request and receive the subnet mask from a router or another host.*



Note

ICMP always reports error messages to the original source.



Note

Important points about ICMP error messages:

- ☐ No ICMP error message will be generated in response to a datagram carrying an ICMP error message.
- ☐ No ICMP error message will be generated for a fragmented datagram that is not the first fragment.
- ☐ No ICMP error message will be generated for a datagram having a multicast address.
- ☐ No ICMP error message will be generated for a datagram having a special address such as 127.0.0.0 or 0.0.0.0.

Thank You